Module - 16:
CI/CD & Automation

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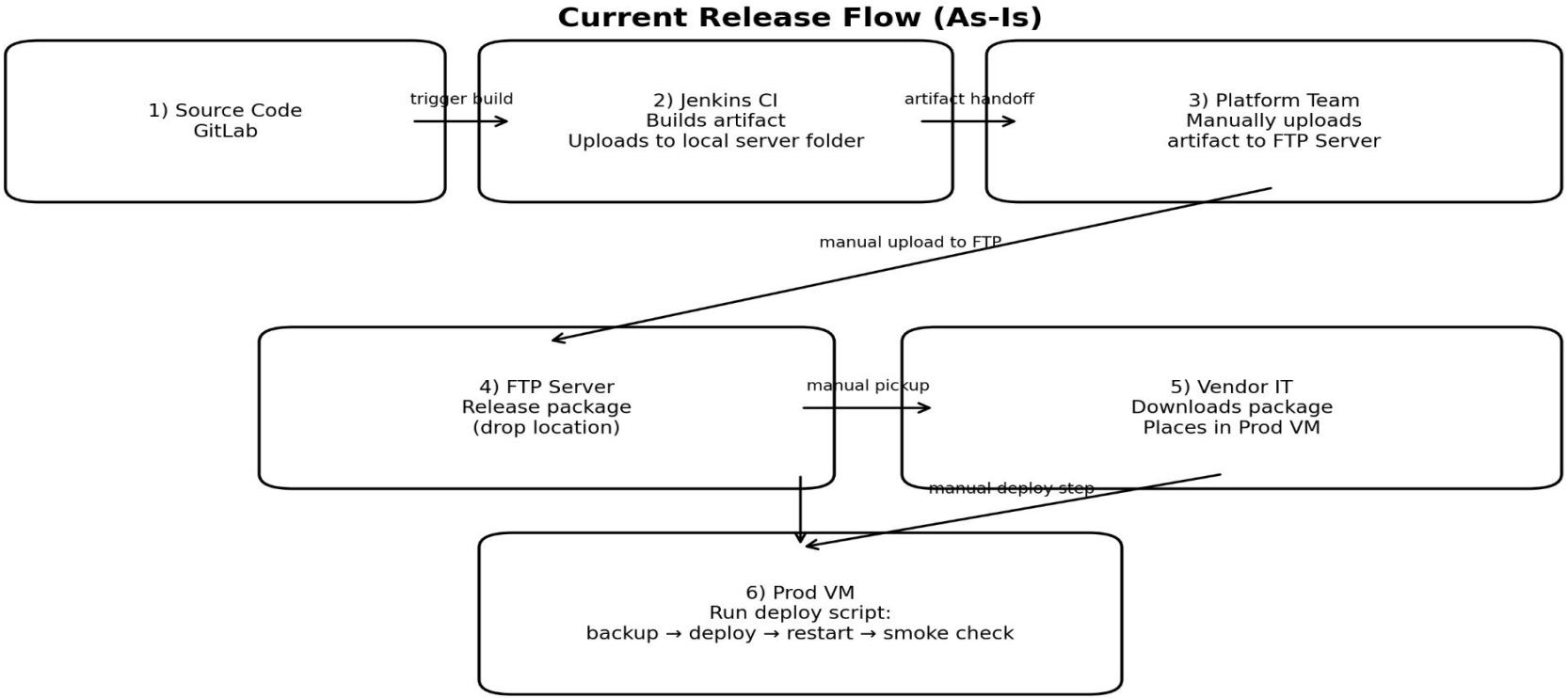
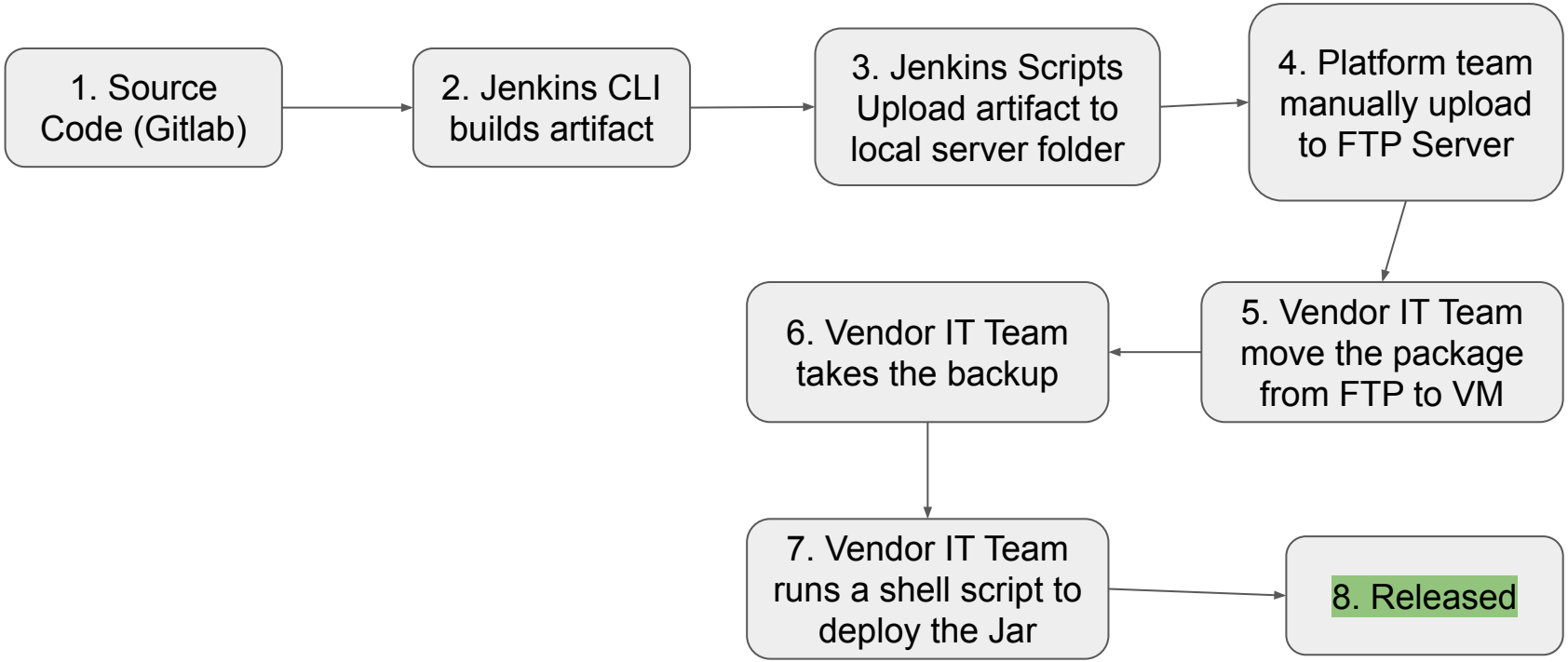
Current Release Flow



Current Release Flow (??)



Questions:

1. How much automation we've right now using Jenkins?
2. How much automation we've in Prod environment using Shell Scripts?
3. How much are we allowed to automate in Production?

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9. Pull Request must find unit tests that are not broken
10. Each month Pull Request ask for extra 10% of unit test coverage (until 95%)

Git Flow to maintain

Branches

- `main` → always releasable, only via PR
- `feature/<ticket>-<short-desc>` → dev branches for new feature
- `bug-fix/<ticket>-<short-desc>` → dev branches for bug fixes
- `hot-fix/<ticket>-<short-desc>` → dev branches for hotfixes

Rules

- No direct commits to `main`
- Everything goes through PR into `main`
- Tag releases from `main` (e.g., `v1.8.0`)

Pull Request rules

PR policy (minimum viable governance)

- Require **≥ 1 approval**
- Require **successful CI pipeline** (nothing broken)
- Require **conversation resolved** (All comment/feedback must be solved)
- Require **up-to-date branch** before merge (no stale PR merges)

Optional (add later)

- Release PR requires one more Approver from Release Manager

Code linter scan

Goal: fail fast on style/quality problems.

Recommended linters (free)

- **Checkstyle** (style rules)
- **SpotBugs** (bug patterns)
- **PMD** (code quality rules)

Blocking rule

- If linter finds violations above threshold → pipeline fails → PR cannot merge.

Unit tests

Requirement

- Unit tests must run on every PR.

Policy

- Test failures should fail the build (tests failing means unsafe),
- but **coverage threshold is not enforced** initially (you can still publish coverage reports).

Implementation

- `mvn test`
- publish JUnit results
- publish coverage as report (non-blocking)

Future Plan:

- Each month increase the required coverage percentage by 10

Upload to FTP Server (automated by pipeline)

Replace the manual “platform team uploads to FTP” step with CI.

FTP directory convention

- `/releases/<app>/<version>/...`
- Example:
  - `/releases/banking-app/1.8.0+105/`
    - `banking-app_1.8.0_105_ab12cd34.zip`
    - `manifest.json`
    - `sha256sum.txt`
    - `release-notes.md`
    - `READY`

Critical control

- Pipeline uploads files first, then uploads `READY` last
- Your existing deployment script (or watcher) only acts when `READY` exists.

Watcher Script in Staging VM

1. A shell Script will check fixed location of FTP Server
2. The Script copy the release package from FTP to Staging VM (safe location)
3. Creates a backup of previous release
4. Unzip the new Release and move to right path
5. Deploy the new release
6. If any exception happens, roll-back to last release automatically
7. Run the script as a service with Cron Job

Thank you